Willian Vieira PhD

Applied modelling and data engineering specialist with a PhD in quantitative ecology. I build mathematical and statistical models using reproducible data workflows that turn complex data into validated, explainable outputs for operational decisions.

DATA SCIENCE & ENGINEER EXPERIENCE

2025 | 2024 (1 yr 9 m)

Data Analyst & developer

Habitat, Montreal, Canada

Built Python/R analytical pipelines and data-processing workflows that turned messy real-world project data into reusable, validated model inputs and decision-ready outputs | Developed metadata-driven ETL, data modelling, and provenance workflows so datasets, definitions, analyses, and handoffs remained traceable, reproducible, and inspectable | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, documentation, and collaborative development practices | Designed cloud/Azure-oriented data infrastructure that made heterogeneous analytical datasets easier to retrieve, validate, and reuse

2024 | 2017

PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Formulated, implemented, and validated mathematical/statistical models for forest dynamics, including Bayesian hierarchical models, simulation workflows, and historical-data evaluation routines for noisy, biased, incomplete data | Ran thousands of simulations on HPC infrastructure () to test algorithmic assumptions, compare model behavior, and scale analyses beyond local compute | Built reproducible pipelines for continental-scale geospatial and environmental data, harmonizing climate, satellite, and field-inventory datasets into modeling-ready inputs | Developed open-source software libraries (,) for demographic modelling with version control, automated testing, documentation, and reusable workflows | Published a technical methods book documenting the full computational pipeline, along with one peer-reviewed publication and two preprints (,)

2022 | 2020 (part-time)

Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec under operational constraints | Led client-facing R&D on spatial bias-correction methods for large-scale ecological monitoring data, translating requirements into a method later adopted by other provinces | Engineered a fully automated and reproducible data pipeline with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

EDUCATION

2024 | 2017

PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations

2016 | 2015

Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

Modelling the dispersion of weed species in agricultural landscapes

2015 | 2010

BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

CONTACT

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w.vieiraw@gmail.com
Linkedin/WillVieira
GitHub/WillVieira
Publications

SKILLS

Mathematical Modelling & Statistics: Bayesian inference · Hierarchical models · Simulation · Optimization · Model validation · Uncertainty

Programming & Database: Python · R · SQL · Stan · Julia · DuckDB · Bash

Data Engineering & Cloud: ETL/ELT · Production ML pipelines · APIs · Azure · AWS · Docker · Apache Arrow · Parquet · HPC

DevOps & Reproducibility: Git · CI/CD · Unix · Unit testing · Make · Quarto · Jupyter · Automated validation

OUTREACH

· Developed and maintained the QCBS R Workshop Series, reaching nearly one thousand graduate students · Taught programming, statistics, SQL, and reproducible workflows to 250+ students across biology, engineering, and graduate programs · Translated stakeholder needs into technical analyses, documented pipelines, and clear analytical outputs

LANGUAGES

Portuguese · Native
French · Full Professional
English · Full Professional